

Lecture 18: Making Magnetic Fields

Pre-Lecture Worksheet

Complete the following worksheet in preparation for the next lecture. The following questions correspond to material covered in chapter 27. Feel free to read chapter 27 and answer the questions. You may also seek answers to the questions from other sources. Note: the textbook will be used as the final source for these questions. While answers gleaned from other sources may be correct, answers must encompass the material covered in the book to be considered correct.

There are three major sources of magnetic fields. For each source:

- (a) Draw the source of the magnetic field
- (b) State what/if anything is moving and label it.
- (c) Draw the orientation of the magnetic field lines.

1) Magnetic Materials

2) Current through a long straight wire

3) Current through a loop or solenoid.

4) How do the number of turns of a loop with a current, effect the strength of the magnetic field produced?

5) (True or false) The strength of the magnetic field for a loop with a current is uniform and strong inside and outside the loop.